Software Test Plan (STP)

Project: Helping Hands (HH), an app that lets management staff of a retail chain volunteer to work in stores during very busy times such as weekends and holidays

Version: 1.0

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1. Introduction

Purpose: This document is the test plan for Helping Hands v1.0. It sets out the objectives, scope, strategy, resources, schedule and responsibilities for testing.

Scope: Testing covers sign-in and access control, stores, busy periods and rules, posting and cancelling shifts, browsing and signing up, eligibility checks, approvals and withdrawals, notifications and reminders, attendance, coverage, reports, audit, and the performance, reliability, usability, accessibility and security requirements. HR, payroll and messaging integrations are excluded because they are not part of this version.

References: Helping Hands SRS v1.0, Helping Hands SAD v1.0, OWASP Testing Guide v4.2, WCAG 2.1.

Definitions: HH (Helping Hands), SRS (Software Requirements Specification), SAD (Software Architecture and Design Specification), RTM (Requirements Traceability Matrix), UAT (User Acceptance Testing), CI (Continuous Integration).

2. Test items

- Auth and RBAC module
- Store, busy period and rules module
- Shift module (post, edit, cancel, list)
- Sign-up module (eligibility, approval, withdrawal)
- Notification module and reminder job
- Attendance module
- Reporting module (coverage, hours, no-shows, CSV)
- Audit log
- REST API (all /api/v1 endpoints)
- Helping Hands web app

3. Features to be tested

Features mapped to SRS requirement IDs:

- HH-F-001: sign in and sign out
- HH-F-002: role and store-scope access
- HH-F-003: manage users
- HH-F-004: manage stores and assign managers
- HH-F-005: define busy periods
- HH-F-006: set volunteering rules
- HH-F-007: post a help request
- HH-F-008: validate shift times and slots
- HH-F-009: edit or cancel a shift and notify volunteers
- HH-F-010: browse and filter open shifts
- HH-F-011: sign up for a shift
- HH-F-012: reject full, cancelled, late or clashing sign-ups
- HH-F-013: weekly hour limit
- HH-F-014: approve or decline without overbooking
- HH-F-015: Filled and Open shift status
- HH-F-016: withdraw before the cutoff
- HH-F-017: in-app notifications
- HH-F-018: 24-hour reminder
- HH-F-019: my schedule
- HH-F-020: record attendance and hours
- HH-F-021: coverage view
- HH-F-022: hours, coverage and no-show reports
- HH-F-023: audit events
- HH-NF-001: shift list and coverage within 2 s
- HH-NF-002: no overbooking under concurrent approvals
- HH-NF-003: data survives restart; daily backup
- HH-NF-004: sign up in three steps on a phone
- HH-NF-005: WCAG 2.1 AA
- HH-NF-006: 70% service test coverage
- HH-NF-007: README setup on a clean machine

Security requirements HH-SR-001 to HH-SR-008 are covered in section 5.1.

4. Features not to be tested

- Pay, overtime or expense handling for volunteered hours (not part of the system).

- HR, payroll, rostering, email, SMS and calendar integrations (future work).
- Whether volunteers really turned up beyond what the store manager records.
- Internals of PostgreSQL, FastAPI and React; only the application's use of them is tested.

5. Test approach / strategy

- **Levels:** unit tests for the services (eligibility, approval counting, cutoffs, hours); integration tests for services with the database, including the approval transaction; system tests of full flows through the web app and API; acceptance tests (UAT) of the main user stories with a store manager and a volunteer.
- **Types:** functional; negative and boundary (last free slot, exactly at the cutoff, back-to-back shifts, weekly limit reached exactly); concurrency; regression on every pull request; performance; security; usability and accessibility on phone and laptop.
- **Entry criteria:** a stable build is deployed to the test environment, test data is loaded, CI runs the unit tests, and the test cases in section 13.1 are reviewed.
- **Exit criteria:** 100% of planned test cases run, no critical or high defect open, no shift ever overbooked in any run, HH-NF-001 to HH-NF-003 met, and the RTM updated.

5.1 Security validation

- Role and store-scope checks: direct API calls outside a user's role or store must get 403 (HH-SR-004).
- Password storage, lockout and session expiry (HH-SR-002, HH-SR-003).
- Input validation and injection attempts on every form and filter (HH-SR-005).
- Contact-detail privacy between volunteers (HH-SR-006) and audit log content (HH-SR-007).
- Rate limiting on sign-in and sign-up (HH-SR-008), TLS configuration (HH-SR-001), and an OWASP ZAP baseline scan.

6. Test environment

Hardware: team laptops for the server and browsers, and at least one Android and one iOS phone.

Software: Helping Hands v1.0 (React web app, FastAPI backend, PostgreSQL 16) run with Docker Compose; Chrome, Edge, Firefox and Safari.

Tools: pytest, Vitest, Playwright, Postman, JMeter, OWASP ZAP, GitHub Actions, Jira for defects.

Test data: a synthetic chain of 5 stores, 5 store managers and 40 volunteers; busy periods for a weekend and the Diwali weekend; shifts with one free slot, full shifts, overlapping and back-to-back shifts, and shifts just before and after the cutoff; 2,000 shifts for the performance test.

7. Test schedule

Testing runs inside the two sprints (Sprint 1: 27-31 Oct 2026, Sprint 2: 02-07 Nov 2026), so each sprint ends with a tested, working product before the presentations from 09 Nov 2026.

Milestone	Date
Test plan approved; CI running unit tests	28-Oct-2026
Test environment and synthetic data ready	29-Oct-2026
Test case design complete (section 13.1)	30-Oct-2026
Sprint 1 testing: sign-in, stores, busy periods, shifts, sign-up rules	29-Oct-2026 to 31-Oct-2026
Sprint 1 review and demo	31-Oct-2026
Sprint 2 testing: approvals, withdrawals, notifications, attendance, reports	03-Nov-2026 to 06-Nov-2026
Concurrency, performance and security testing	05-Nov-2026 to 06-Nov-2026
Regression run and UAT	07-Nov-2026
Test summary report and RTM update	08-Nov-2026

8. Test deliverables

- Test plan (this document)
- Test cases, manual and automated (section 13.1)
- Test scripts (pytest, Playwright, Postman collections, JMeter plans)
- Test data generation script
- Test execution logs and CI results
- Defect reports (Jira)
- Updated RTM
- Test summary report

9. Roles and responsibilities

Role	Name	Responsibility
Scrum Master / QA Lead	Dhanush Shankar Reddy (PES1UG24AM083)	Run the Jira board and sprints, track the burndown, prepare this plan, coordinate test runs and report status
Product Owner	Aamer Khan (PES1UG24AM902)	Prioritise the backlog, accept finished stories, approve test results, run UAT and sign off
Developer	Vivek Varma (PES1UG24AM070)	Unit tests, CI pipeline, defect fixes and triage
Test Engineer	Bhuvan Kumar P (PES1UG24AM069)	Design and run test cases, automate regression, run the security and performance tests, log defects

10. Risks and mitigation

Risk	Mitigation
Delay in a stable build	Ask for early smoke builds; test finished modules first
Overbooking only shows under concurrency	Automated test firing two approvals for the last slot at once
Test data misses time boundaries	Include shifts exactly at the cutoff, back-to-back shifts and the weekly limit reached exactly
Test environment downtime	Docker Compose setup that runs on any team laptop

Risk	Mitigation
Little time for performance and security testing	Fixed window on 05 and 06 Nov; ZAP scan automated in CI

11. Assumptions & dependencies

- Requirements in SRS v1.0 are frozen before test case design.
- Synthetic test data is accepted as representative of a real chain.
- A store manager and a volunteer from the team are available for UAT.
- All four team members can run the Docker environment and the CI pipeline.

12. Suspension & resumption criteria

Suspend testing if:

- The test environment is unavailable for more than 4 hours.
- The build is too unstable and blocks more than 30% of test cases.
- A critical integrity or security defect is found, such as an overbooked shift or a cross-store change; integrity and security tests pause until it is fixed.

Resume testing if:

- Blocking defects are fixed and a new build passes the smoke test.
- The environment is stable again.
- Affected test cases are run again from the start.

13. Test case management & traceability

The RTM (SRS section 8) maps each SRS requirement to its test cases. For example:

- HH-F-001 (sign in) → TC-AUTH-01
- HH-F-012 (reject ineligible sign-up) → TC-SIGN-02 to TC-SIGN-05
- HH-F-014 (approve without overbooking) → TC-APPR-01, TC-CONC-01
- HH-NF-001 (shift list response time) → TC-PERF-01
- HH-SR-004 (server-side access checks) → TC-AUTH-03, TC-AUTH-04

13.1 Test cases

TC ID	Req ID	Title	Steps / Input	Expected result	Type
TC-AUTH-01	HH-F-001	Sign in	Sign in as a volunteer with a valid email and password	Session starts and the volunteer home opens	Functional
TC-AUTH-02	HH-F-001	Wrong password	Sign in with a wrong password	General error shown; no session created	Functional
TC-AUTH-03	HH-F-002, HH-SR-004	Role check	As a volunteer, call POST /api/v1/shifts directly	403; nothing saved	Security
TC-AUTH-04	HH-F-002, HH-SR-004	Store scope	As manager of store A, edit a shift of store B	403; shift unchanged	Security

TC ID	Req ID	Title	Steps / Input	Expected result	Type
TC-USER-01	HH-F-003	Create user	Admin creates a store manager account	Account is active with the Store Manager role	Functional
TC-STORE-01	HH-F-004	Create store	Admin creates a store and assigns its manager	Store listed with its manager	Functional
TC-BP-01	HH-F-005	Busy period	Admin adds "Diwali weekend", 07-11-2026 to 08-11-2026	Busy period can be chosen when posting shifts	Functional
TC-RULE-01	HH-F-006	Volunteering rules	Admin sets the weekly limit to 8 hours	New sign-ups are checked against 8 hours	Functional
TC-SHIFT-01	HH-F-007	Post shift	Post a billing shift on 08-11-2026, 10:00 to 14:00, needing 3 volunteers	Shift is Open with 3 free slots	Functional
TC-SHIFT-02	HH-F-008	End before start	Post a shift from 14:00 to 10:00	Validation error; nothing saved	Boundary
TC-SHIFT-03	HH-F-008	Zero volunteers	Post a shift needing 0 volunteers	Validation error; nothing saved	Boundary
TC-SHIFT-04	HH-F-009, HH-F-017	Cancel shift	Cancel a shift with 2 confirmed volunteers	Shift Cancelled; both volunteers notified	Functional
TC-BROWSE-01	HH-F-010	Filter shifts	Filter open shifts by busy period and store	Only matching shifts, with correct free slots	Functional
TC-SIGN-01	HH-F-011	Sign up	Volunteer signs up for an open shift	Sign-up Requested; manager notified	Functional
TC-SIGN-02	HH-F-012	Full shift	Sign up for a shift with no free slots	409 SHIFT_FULL	Negative
TC-SIGN-03	HH-F-012	Clash	Holding 10:00 to 14:00, sign up for 13:00 to 17:00 the same day	409 CLASH, naming the other shift	Negative
TC-SIGN-04	HH-F-012	Back-to-back	Holding 10:00 to 14:00, sign up for 14:00 to 18:00	Allowed; touching shifts do not clash	Boundary
TC-SIGN-05	HH-F-012	Short notice	Sign up 2 hours before start (notice rule 12 hours)	409 NOTICE	Boundary
TC-SIGN-06	HH-F-013	Weekly limit	With 6 confirmed hours, sign up for a 4-hour shift (limit 8)	409 WEEKLY_LIMIT; 2 hours left shown	Boundary
TC-APPR-01	HH-F-014, HH-F-015	Approve last slot	Approve a sign-up when 1 slot is free	Confirmed; shift becomes Filled	Functional
TC-APPR-02	HH-F-014	Approve on full	Approve a sign-up on a Filled shift	409; confirmed count unchanged	Negative
TC-APPR-03	HH-F-014	Decline	Decline a sign-up	Declined; volunteer notified	Functional
TC-CONC-01	HH-F-014, HH-NF-002	Last-slot race	Two managers approve different sign-ups for the last slot at the same moment	Exactly one approval succeeds	Concurrency
TC-STAT-01	HH-F-015	Slot reopens	A confirmed volunteer withdraws from a Filled shift	Shift goes back to Open	Functional

TC ID	Req ID	Title	Steps / Input	Expected result	Type
TC-WD-01	HH-F-016	Withdraw in time	Withdraw 30 hours before start (cutoff 24 hours)	Withdrawn; slot freed; manager notified	Functional
TC-WD-02	HH-F-016	Withdraw late	Withdraw 3 hours before start	409; sign-up stays Confirmed	Boundary
TC-NOTE-01	HH-F-017	Notifications	Approve, decline and cancel one sign-up each	One notification per event, to the right user	Functional
TC-NOTE-02	HH-F-018	Reminder	Confirmed shift starts in 24 hours; run the reminder job twice	Exactly one reminder sent	Functional
TC-SCHED-01	HH-F-019	My schedule	Volunteer opens My Schedule	Only their own shifts, with status and hours	Functional
TC-ATT-01	HH-F-020	Record attendance	Mark a volunteer attended with 4 hours	Saved; hours in the volunteer's history	Functional
TC-ATT-02	HH-F-020	Invalid hours	Enter 6 hours for a 4-hour shift	Validation error; nothing saved	Boundary
TC-COV-01	HH-F-021	Coverage view	Open coverage for "Diwali weekend"	Needed, requested and confirmed match the data	Functional
TC-REP-01	HH-F-022	Hours report	Hours by volunteer for one busy period, then export CSV	Totals match attendance; CSV opens	Functional
TC-AUDIT-01	HH-F-023, HH-SR-007	Audit entry	Approve a sign-up, then open the audit history	Entry shows action, time and manager; no secrets	Functional / Security
TC-PERF-01	HH-NF-001	Response time	JMeter: 50 users load the shift list (2,000 shifts) 20+ times each	90th percentile under 2 s	Performance
TC-REL-01	HH-NF-003	Restart	Commit shifts and sign-ups, then restart the application	All committed records intact	Reliability
TC-UX-01	HH-NF-004	Phone sign-up	On a 360 px phone screen, sign up from the home page	Done in 3 steps or fewer	Usability
TC-UX-02	HH-NF-005	Accessibility	Keyboard-only sign-up and approval; automated WCAG check	All controls reachable; no AA violations	Accessibility
TC-SEC-01	HH-SR-002	Password storage	Inspect the users table and the logs	Only salted hashes; no plaintext	Security
TC-SEC-02	HH-SR-002	Lockout	Enter a wrong password 5 times	Account locked for 15 minutes	Security
TC-SEC-03	HH-SR-003	Session expiry	Stay idle 30 minutes, then try an action	Asked to sign in again	Security
TC-SEC-04	HH-SR-005	Injection attempt	Search shifts for: ' OR '1'='1	Treated as text; no data or error leaked	Security
TC-SEC-05	HH-SR-006	Contact privacy	Volunteer B opens a shift where volunteer A is confirmed	A's phone and email not shown	Security
TC-SEC-06	HH-SR-008	Rate limit	Send 30 sign-up requests in one minute	Requests over the limit get 429	Security

TC ID	Req ID	Title	Steps / Input	Expected result	Type
TC-SEC-07	HH-SR-001	TLS only	Connect to the deployed server over HTTP and over TLS 1.0	HTTP redirected; TLS 1.0 refused	Security
TC-DEV-01	HH-NF-006	Unit tests	Run the service unit tests in CI	All pass; line coverage at least 70%	Maintainability
TC-DEV-02	HH-NF-007	Clean setup	Fresh clone; follow the README on a clean machine	App starts; demo accounts work	Portability

14. Test metrics & reporting

Metrics collected:

- % of test cases executed
- % passed, failed and blocked
- Defect density per module
- Defect ageing and severity distribution
- Requirement coverage (requirements with at least one passed test)
- Unit test line coverage of the services

Reports:

- Daily execution status (Jira dashboard)
- Weekly defect trend
- Final test summary report

15. Approvals

Role	Name	Signature / Date
QA Lead	Dhanush Shankar Reddy (PES1UG24AM083)	
Dev Lead	Vivek Varma (PES1UG24AM070)	
Product Owner	Aamer Khan (PES1UG24AM902)	